Manuel Oelmaier

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Profile

Software engineer focused on LLM systems: research, evaluation, and deployment. Working student at Agile Robots, owning the agent harness for an internal tool (tools, UI, eval, safeguards). B.Sc. Computer Science, TUM (coursework grade 2.7; thesis still being graded). M.Sc. Robotics, Cognition, Intelligence from Oct. 2026, expected 2028 (AI/ML). Thesis on token eviction in linear-attention transformers; code-generation finetuning with a public eval harness; analysis infrastructure for the FIDE Candidates 2026.

Education

Master of Science – Robotics, Cognition, Intelligence Oct. 2026 – expected 2028
Technical University of Munich, Germany

- Focus: AI, machine learning, intelligent systems

Bachelor of Science in Computer Science 2021 – Sep. 2026
Technical University of Munich, Germany

- Coursework grade 2.7; thesis submitted, still being graded
- Focus: machine learning, artificial intelligence, software development
- Relevant coursework: Praktikum Next Gen AI, Ethics in Artificial Intelligence
- *Thesis:* **Hidden State Informed Token Eviction**

Professional Experience

Working Student – Agile Robots SE Sep. 2026 – present
Munich, 20 h/week. Own the implementation of the agent harness for an internal tool.

- Building its tools, UI, evaluation, and safeguards

Infrastructure Engineer – Team Sindarov / FIDE Candidates 2026 2025 – 2026
Project role, remote and on-site. One of two engineers behind the distributed engine-analysis infrastructure used throughout preparation for and during the FIDE Candidates Tournament 2026. Owned setup, deployment, and day-to-day technical communication with players and seconds.

- Co-owned hardware selection and performance tuning to maximize sustained engine throughput
- Access-controlled remote analysis for confidential opening preparation
- Built an internal observability and control dashboard: live status, cost-aware on-demand scaling, audit logs
- Primary technical point of contact: onboarding, training, troubleshooting, feature requests

IONOS Partner Award, AI Project of the Year 2026, with IS4IT at the IONOS Summit in Berlin, for this platform. Zero unplanned downtime.
Team Sindarov won the Candidates 10/14, the best result in the tournament's history.

Projects

Bachelor thesis: Hidden State Informed Token Eviction submitted; grading in progress

- Efficient LLM inference: learnable KV-cache token eviction routed by a linear-attention memory's reconstruction-error signal
- Benchmark harness with accuracy/throughput ablations; hands-on PyTorch implementation and evaluation
- manueloelmaier.de/blog/efficient-llm-inference-thesis

Praktikum Next Gen AI: LLM finetuning & VS Code extension 2025

- Finetuned transformer models for Python code generation; built an automated multi-case evaluation pipeline
- Raised the score from 40 % to 75 % on that multi-case code-generation eval
- Published the best model as a VS Code extension for local, privacy-preserving in-editor queries
- github.com/manuel-Oelmaier/LocalPythonCodingLlm

Self-hosted AI platform – Homelab 2025 – ongoing

- Own-hardware stack (Docker, Ollama, LibreChat, MCP agents); RAG over local documents, injected into agent and chat flows
- Compared retrieval against putting the same documents straight into the prompt, on consumer hardware
- manueloelmaier.de/blog/homelab-ai-platform

Skills

Technical: Python, TypeScript, Bash, SQL; also Java, C, MongoDB
AI / ML: PyTorch, Hugging Face, LLM finetuning & evaluation, sparse attention, KV-cache eviction, RAG, MCP
Systems / Infra: Linux, Git, Docker, distributed/HPC workloads, performance tuning (CPU affinity, NUMA), SSH & network security, observability dashboards, cost-aware cloud scaling
Languages: German (native), English (full professional proficiency, C1; English-taught M.Sc.)

Volunteering & Hobbies

- Volunteer youth chess trainer, SC Eching
- Competitive tournament chess player

References

IM Roman Vidonyak – Chief Trainer, Team Sindarov (winner of the FIDE Candidates 2026 and challenger for the World Chess Championship); Head Trainer, FC Bayern Munich Chess
Introduction and recommendation letter on request.